MOHAR A WALKTHROUGH FOR JUDGES

How we prove who held the exam paper.

Fifteen steps, in plain language. Every technical word is explained the first time it appears — nothing here assumes you already know it.

What this is · a custody record for physical question papers
Where it runs · the exam centre itself, with no internet
Navigate · ↓ ↑ or scroll

01 THE PROBLEM

Papers don't leak because someone breaks in.

They leak because a person who was *allowed*
to hold the paper handed it to somebody else.

In the Hazaribagh case the school principal was authorised to be in the strong room. No lock was broken. What was missing was any record of who had the package and when — the register was written up afterwards, by hand, and it said whatever was convenient.

So the real question is not "how do we stop a leak"

Nothing can promise that. The question is: **when a leak happens, can we prove who held the paper at every moment — and can that proof survive the people who would most like to change it?**

02 WHAT WE BUILT

A register nobody can edit. Including us.

Every step of the paper's journey is written down as a signed record, in a chain, where nothing can be changed or removed afterwards.

Signed

Each device has its own secret key. Anyone can later check which device wrote a record — and nobody can forge one.

Chained

Each record carries a fingerprint of the one before it. Alter an old record and every record after it stops matching.

Append-only

The database gives our software no permission to edit or delete. That is not a promise in code — it is a permission that does not exist.

03 THE JOURNEY

The paper's route is the thing we're recording.

- **Printing press**
  The paper is encrypted and sealed
- **Dispatch**
  Handed to a courier — two signatures
- **In transit**
  Checkpoint scans along the road
- **Overnight custody**
  Received by the custodian
- **The exam centre**
  Delivered to the superintendent
- **The unlock**
  Two officials, minutes before the exam

Most of the risk is not in the strong room. It's in the hours the package spends in a vehicle and in somebody's custody overnight.

That stretch is exactly what a paper register describes worst — so it is the stretch we record most carefully. Every hop below produces its own signed record.

04 AT THE PRESS

The paper is locked into a box only a key can open.

The question paper is scrambled into unreadable data. Without the key, it is not "hard to read" — it is nothing at all.

THE TECHNICAL NAME `XChaCha20-Poly1305`

An encryption method that also detects tampering. If a single bit of the sealed data is altered, it refuses to open rather than producing plausible-looking rubbish — which matters when the alternative is handing out a wrong paper an hour before an exam.

```
Original paper
↓ encrypt
a8f3c1d9e0b4… ← unreadable
```

05 AT THE PRESS

The key is broken into four pieces. Any three open it.

No single person ever holds the whole key —
so no single person can open the paper alone.

Two pieces tell you **nothing**. Not "almost enough" — mathematically nothing, as if you had none at all.

THE TECHNICAL NAME **Shamir Secret Sharing (3 of 4)**

A way of splitting a secret so that a set number of pieces rebuilds it and fewer reveal nothing. Four pieces mean one can be lost without locking everyone out.

Piece 1

The exam authority

Piece 2

Time-locked — see next slide

Piece 3

The superintendent

Piece 4

The observer

2 PIECES → NOTHING

3 PIECES → KEY REBUILT

06 AT THE PRESS

One piece doesn't exist yet.

The second piece cannot be stolen, leaked, or demanded — because until the exam hour there is nothing there to take.

It becomes available on a public schedule that nobody can hurry, not even us. This is the one part of the system that **prevents** rather than merely records.

THE TECHNICAL NAME `timelock encryption (drand)`

A public service publishes an unpredictable value on a fixed clock. Data can be locked to a future publication and cannot be opened before it arrives.

WHY THIS IS THE STRONGEST CLAIM WE MAKE

Every other control here is **detective** — it records what happened so it cannot be denied later. This one is **preventive**: early release is not merely detectable, it is impossible, and it is mathematics doing the work rather than policy.

07 THROUGHOUT

A register where tearing out a page is obvious.

Every record includes a summary of the record before it. Change an old entry and every entry after it stops matching.

THE TECHNICAL NAME **hash chain · SHA-256 · Ed25519**

A *hash* is a short fingerprint of data — change one letter and the fingerprint changes completely. A *signature* proves which device wrote it. Together: you can tell what was written, by whom, and in what order.

`Record 1001` → `hash A`

↓

`Record 1002` carries `hash A` → `hash B`

↓

`Record 1003` carries `hash B` → `hash C`

This is not a blockchain. A blockchain exists so that many parties who distrust each other can agree on a shared record. Here there is one exam board — the risk is people inside it rewriting history, and that is solved by signatures and database permissions, without a network of miners.

08 ON THE ROAD

Every handover needs two signatures.

The person handing over and the person receiving both sign – on two different devices. One device signing twice is one person's word.

A photograph is taken at each handover and its fingerprint recorded, so the record cannot later be attached to a different moment.

THE TECHNICAL NAME `co-signature requirement`

The ledger refuses a handover record that carries only one signature, or two signatures from the same device.

District officer → Courier
↓ checkpoint scan
↓ checkpoint scan
Courier → Custodian
↓
Custodian → Centre

09 AT THE CENTRE

A small box, bolted to the wall, that never sleeps.

It reads fingerprints, keeps its own clock, and reports in every thirty seconds — with no internet at all.

The absence of a report is itself recorded. Unplugging the device is not a way to go dark; it is a way to raise an alarm.

THE TECHNICAL NAME `heartbeat · store-and-forward`

Records are written down on the device before any attempt to send them. Cutting the power delays a record; it does not erase one.

Fingerprint reader

Matches on its own chip and reports only a slot number and a confidence score.

Battery-backed clock

Correct timestamps with no power and no network.

Its own secret key

Signs every record. The key never leaves the chip.

Buzzer

Sounds when access is refused, so a refusal is not silent.

10 AT THE CENTRE

We never store anyone's fingerprint.

The reader compares the finger inside its own chip and tells us only *"slot 3 matched, score 187"*. No image, no template, ever leaves it.

So a breach of our database cannot leak a fingerprint — because there is no fingerprint in it to leak.

THE TECHNICAL NAME **on-module matching · data minimisation**

Under India's DPDP Act 2023 biometric data is personal data. The cleanest position to be in is not to hold it.

Finger → matched inside the reader

↓

"slot 3, score 187"

↓

who slot 3 is → a separate register

11 THE UNLOCK

Two officials. Two minutes. Two different fingers.

The first official touches the reader. The second has two minutes to follow — and must be a different person.

One person tapping twice is refused and recorded as *same finger twice*. Nobody arriving is recorded as *window expired*. Both are kept.

THE TECHNICAL NAME `two-person co-presence window`

The rule checks two distinct enrolled fingers inside 120 seconds. A photograph is taken at each match and its fingerprint recorded.

TWO_PERSON_CONFIRMED

SAME_FINGER_TWICE

WINDOW_EXPIRED

All three are recorded

A refusal is evidence too. The system keeps the attempts that failed, not only the ones that worked.

12 THE UNLOCK

Twenty-one checks. Every one of them, every time.

The answer starts at *no*. Each check must positively pass. We never stop at the first failure — an attempt that trips four checks is a different story from one that trips a clock error.

key presented	key valid
key in window	key stage match
device enrolled	device binding
roster membership	role permitted
geofence	position accuracy
clock skew	custody window
seal serial	package state
exam active	first fingerprint
second fingerprint	two-person window
photograph taken	seal lock — no sensor
room occupancy — no sensor	

THE TECHNICAL NAME **deny by default · no short-circuit**

The attempt is written to the record *before* the answer is returned — so there is no sequence of events in which someone learns the outcome without the attempt being on record.

THE TWO GREY ONES

Those two checks report **"not evaluated"** because we have no seal-lock sensor and no occupancy sensor on the bench. The system does not quietly pass them. A demonstration that scored 21 out of 21 would be telling you something untrue.

13 THE UNLOCK

Only now does the paper open.

ACCESS GRANTED

↓

3 of 4 pieces → key rebuilt

↓

sealed data → the original paper

The key is rebuilt from three pieces, the sealed data is opened, and the paper is readable for the first time since the press.

And we check one more thing: that the paper which came out matches the fingerprint recorded when it went in. So it is not merely *a* document that opened — it is **the** document that was sealed.

14 WHEN IT GOES WRONG

A refusal is the most important record we keep.

Wrong key, wrong person, wrong place, wrong hour — the attempt is refused, the buzzer sounds, and a photograph is taken of whoever is standing at the terminal.

It appears in the control room immediately, on its own page, with the exact reason and the photograph's fingerprint committed to the chain.

THE TECHNICAL NAME **Server-Sent Events**

A one-way live feed from the server to the screen. We chose it over a two-way connection because the screen only ever listens — and this way it reconnects by itself when the network drops.

refused → buzzer
↓ photograph taken
↓ fingerprint of the photo → chain
↓ control room, live

What this cannot do.

If a person reads the paper and memorises it, no custody system on earth will stop that — and ours doesn't claim to.

What we can do is narrow it: fewer people see the paper, for less time, and if it leaks the investigation starts with **two named people** instead of five hundred.

Built and demonstrable

Encryption, key splitting, signatures, the chain, the 21 checks, the fingerprint ceremony, the live control room.

Designed, not yet built

Hardware attestation of devices, external timestamping of the daily summary, and the seal-lock and occupancy sensors.

Simulated on the bench

Where a sensor isn't fitted, the screen says so. Nothing is shown as passing that we could not measure.

> We didn't come to change how an exam board works. We came to give them a way to prove they did it correctly.

↓ EVERY WORD, IN ONE PLACE

The vocabulary.

TERM	WHAT IT MEANS HERE
hash	A short fingerprint of data. Change one letter and the fingerprint changes completely.
SHA-256	The specific fingerprinting method we use.
signature	Proof that a particular device wrote a record, and that it hasn't been altered since.
Ed25519	The signing method. Chosen because it needs no randomness at signing time — a weak random number on a cheap chip can leak the secret key.
hash chain	Records linked so each carries the fingerprint of the previous one. Tampering with any of them breaks all the ones after.
append-only	Records can be added but never edited or deleted — enforced by the database, not by our code.
encryption (AEAD)	Scrambling data so it's unreadable without the key, in a way that also detects tampering.
Shamir Secret Sharing	Splitting a key into pieces so a set number rebuilds it and fewer reveal nothing.
timelock	Locking data so it cannot be opened before a specific future moment.
geofence	A check that the request came from inside the centre's boundary.
custody epoch	A key that only works for a six-hour window. If it leaks, it stops working on its own.
heartbeat	A regular "I'm still here" message. Its absence is recorded as an event.
store-and-forward	Writing a record down locally first, then sending it when the network allows.
deny by default	The answer starts at no; every check must positively pass to change it.
Server-Sent Events	A live one-way feed from server to screen, so the control room updates without refreshing.